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Machine Learning in IT Service Management

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Abstract

IT Service Management (ITSM) is a variety of activities directed towards maintenance of IT infrastructure. Hence, it is considered to be an important activity for any company, even to one not related to IT. Time of incidents' resolution is the key performance indicator for ITSM. To reduce resolution time, authors propose infrastructure incident prediction model. Model is based on machine learning technologies. Application of machine learning models in ITSM allows significant improves in customer experience and handling issues more efficiently, decreasing service desk agents' efforts and reducing service costs.
This paper aims to propose predictive method of the incident resolution time estimation. Proposed model derives insights from incident data and predicts estimated time of resolution, allowing detection of incident resolution's de-lays.
Authors analyzed prediction accuracy, and derived results demonstrating that model can assist with prediction for a large set of incidents using dataset based on a real service desk incident data. Additionally, its practical use is applicable for service improvement process.

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Keywords: machine learning, information technology services management, gradient boosting decision trees

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1. Introduction

IT Service Management (ITSM) is a variety of activities directed towards maintenance of IT infrastructure. Hence, it is an important activity for any company, even to one not related to IT [1]. Incident management (IM) is an ITSM process area. The first goal of the IM process is to restore a normal service operation as quickly as possible and to minimize the impact on business operations, thus ensuring that the best possible levels of service quality and availability are maintained. Normal service operation is defined here as service operation within service-level agreement (SLA). SLA is a document that describes the performance criteria a provider promises to meet while delivering a service [2]. This agreement also sets out the remedial actions and any penalties that will take effect if performance falls below the promised standard. It is an essential component of the legal contract between a service consumer and the provider. Time of incidents' resolution is the key performance indicator for IM. Ticket management system store lots of data: ticket creation time, SLA criterias, assigned specialist, ticket priority and its impact, etc. Next in this paper we will assume that SLA is time agreed for the ticket to be solved. It gives us more chances to predict potential SLA violation in such condition.

In this paper we use dataset of ITSM of fast food restaurant chain. The objective is to develop a predictive model, that can help improve quality of IT service.

The evaluation method is F1 score. It's a harmonic mean of the precision and recall rate. Prediction and recall are the common metrics used in classification. F1 score will take into account both the precision and recall.

The paper is organized as following. Section 2 introduces features generated through analysis, Section 3 describes further features' analysis and compares sampling techniques. Section 4 introduces the model that is used to generate the final predictions. Paper is concluded with future works direction.

2. Feature engineering

Feature engineering is an essential part of building of any intelligent system, and this process is both difficult and expensive. To obtain the training dataset, we first pre-process incoming raw data: filter not closed tickets, transform duration fields from string format to integer (they represent duration in minutes), convert date-time string to date-time object and encode categorical fields to integers.

To label a training instance, we use "1" to represent the ticket SLA was violated, this case is called positive instance; "0" represents the situation when ticket SLA was not violated, and we note this case as a negative instance.

The feature engineering is about the patterns of ticket, specialist, ticket's type individually and the combination of the interaction patterns between two of the them [3]. The architecture of the features is below:

- Ticket feature:
  - Derived features
- Specialist feature:
  - Basic aggregated features
  - Probability of violation
- Category feature:
  - Basic aggregated features
  - Probability of violation
- Category-Specialist feature:
  - Basic aggregated features
  - Probability of violation

To make the prediction precise, it's very important to depict a ticket thoroughly [4,5]. To do it well, we studied cleaned data and then found that week information and time of creation are very important for the final result. For example, most of the tickets which were created at weekend were resolved with violations.

After additional analysis we listed the following key features:

- Ticket creation day of week
- Ticket creation day is day off

- Ticket creation hour
- Ticket SLA deadline day of week
- Ticket SLA deadline day is day off
- Ticket SLA deadline hour

The features are summarized in 2 aspects: the basic aggregated features and their ratio as probabilities. The raw count of the behaviors in a selected set of periods could provide the information about the work pattern of a particular specialist. It is also a basic preparation stage for the other complex features that will be used in our analysis. So, let's describe features as the following:

- Specialist resolved tickets total count in last n days:

$$W_s = \sum_{d \in Dates} \sum_{t \in Tickets_s} 1 \quad (1)$$

where Dates is a set of last n days and $Tickets_s$ is a set of tickets in day d closed by specialist s.

- Specialist SLA violated tickets total count in last n days:

$$VW_s = \sum_{d \in Dates} \sum_{t \in Tickets_s} isViolated(t) \quad (2)$$

where isViolated is a Boolean function indicating that the ticket t resolution time violated the SLA, Dates is a set of last n days and $Tickets_s$ is a set of tickets in day d resolved by specialist s.

- Hence, probability of SLA violation in last n days:

$$P_s = \frac{VW_s}{W_s} \quad (3)$$

Besides of the specialist's and ticket's features, listed above, raw average of resolve duration and its ratio to SLA in a selected set of categories could provide the information about the pattern of the ticket.

- Average tickets' resolution time of category in last n days:

$$A_c = \frac{\sum_{d \in Dates} \sum_{t \in Tickets_{d,c}} duration_t}{\sum_{d \in Dates} \sum_{t \in Tickets_{d,c}} 1} \quad (4)$$

where $duration_t$ is a duration of resolution of ticket t in minutes, Dates is a set of last n days and $Tickets_{d,c}$ is a set of tickets in day d in category c.

- SLA duration to average tickets' resolution time of category ratio in last n days:

$$R_c = \frac{Ac}{SLA_c} \quad (5)$$

where SLA_c is ticket SLA duration for ticket in category c in minutes.

The consideration of specialist-category's features is similar to specialist's. In the following section, we will first assume that basic aggregated features have been extracted and then describe the details of features that are mostly special and important to the prediction problem.

- Specialist resolved tickets total count of category in last n days:

$$W_{s,c} = \sum_{d \in Dates} \sum_{t \in Tickets_{s,c}} 1 \quad (6)$$

where Dates is a set of last n days and $Tickets_{s,c}$ is a set of tickets in day d of category c closed by specialist s.

- Specialist SLA violated tickets total count of category in last n days:

$$VW_{s,c} = \sum_{d \in Dates} \sum_{t \in Tickets_{s,c}} isViolated(t) \quad (7)$$

where isViolated is a Boolean function indicating that the ticket t resolution duration violated SLA, Dates is a set of last n days and $Tickets_{s,c}$ is a set of tickets in day d of category c closed by specialist s.

- Probability of SLA violation of category in last n days:

$$P_{s,c} = \frac{VW_{s,c}}{W_{s,c}} \quad (8)$$

3. Feature analysis and selection

Feature analysis is another key issue before applying learning algorithms. The central premise when using a feature selection technique is that the data contains many features that are either redundant or irrelevant, and can thus be removed without incurring much loss of information. As we know more about decision tree model, we start to learn the distribution of the values among features in probability theory. The tools provided by Pandas could provide such distribution information.

Firstly, we extract features to each day in data, then use last two days for test and the other days for train. For whole data set, we further divide it into positive data set and negative data set. Then, we calculate some statistical quantity, like min/max value, mean, variance, skewness etc. Secondly, we compare the values between positive and negative data set, and then followed by comparison be-tween train and test data set in each of positive and negative data set. The former experiment can provide information about how clear the separation between positive and negative is.

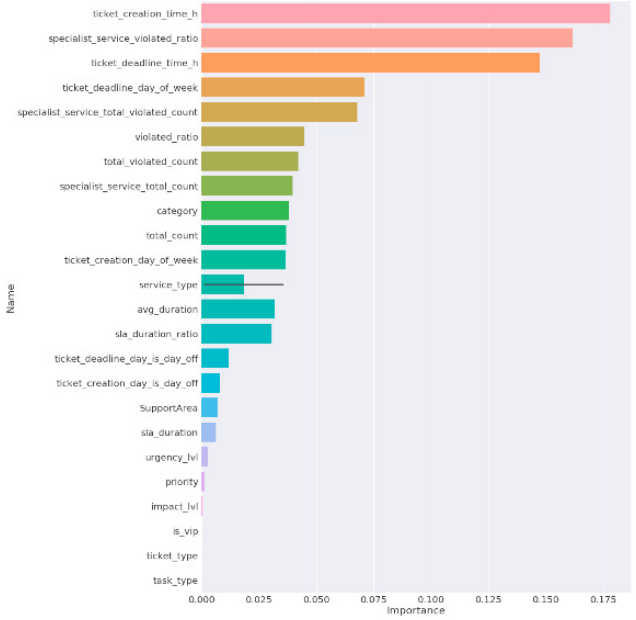


Fig. 1. Factors importance chart.

Building factors importance chart helps us in visualizing all SLA failure influencing factors and finding anomalies. Beside simple statistical analysis of feature we use random forests for feature selection. Random forests are among the most popular machine learning methods thanks to their relatively good accuracy, robustness and ease of use. They also provide two straightforward methods for feature selection: mean decrease impurity and mean decrease accuracy. Most features mentioned above, except “impact level”, “VIP indicator”, “Ticket type” and “Task Type”, are stable in this type of test.

4. Model

During the period of the development, we have explored many different learning models, like Naive Bayesian classifier, Logistic Regression and Gradient Boosting Decision Trees model [6]. In our experiments, Gradient Boosting Decision Trees model perform much better than others, so only the GBDT model is selected as our final model. In the following section, we will introduce the model we used and parameter tuning of the model. Gradient Boosting Decision Trees is a hot topic among data competition in recent years. GBDT is a tree-based boosting learning algorithm [7]. We use the classifier provided in Sklearn library. The optimization goal of GBDT is to optimize deviance which is same as logistic regression. Unlike Random Forests, GBDT combines different tree estimators in a boosting way. That means a GBDT model is built sequentially by using weak decision tree learners on reweighed data learning the deviance. Then it sums up the prediction score of built trees with a given learning rate to generate a powerful learner. We use the following parameters to train the gradient boosting trees:

- Learning rate: 0.5
- Maximum tree depth: 32
- Minimal leaf size: 1
- Minimal samples split: 0.1
- Number of estimators: 60
- Subsample: 1.0
- Number of features: $\sqrt{N}$

The first issue is unbalanced label ratio of the training data. The positives only account for 9.86% of the total records. To fix this problem, we try 2 sampling strategies. The first is random under sampling of the training part of data set. The second is SMOTE oversampling with equal classes sizes. The experiment shows that the latter is better and is chosen as the final sampling method. Prediction result is evaluated in F1 score and the score is 82.39% with precision as 82.32% and recall as 82.47%.

5. Conclusion

In this paper, we presented our prediction model for SLA violation in ITSM of fast food restaurant chain. The paper focus on the feature engineering and model engineering. In the feature engineering section, we propose a set of basic aggregated features and some not obvious derived features. The derived features dramatically improve the overall models and could be further used in other prediction cases. We also use advanced oversampling technique to improve the models. It improves the diversity of models and result is evaluated in a more comprehensive manner. The latter improve the accuracy of the prediction.

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